

Annex B

Cover Page for College to submit with Prelim Examination

Paper

ACJC

Qn/No	Topic Set	Paper No.	Answers	Marks
1	Inequalities	9740/01	$1 - \sqrt{6} \leq x < 0$ or $x \geq 1 + \sqrt{6}$; $x < -(1 + \sqrt{6})$ or $x > 1 + \sqrt{6}$	5
2	Sigma Notation	9740/01	$e^{\frac{1}{2}(n-1)(n+4)}$	5
3	Equations	9740/01	\$1497	4
4	Mathematical Induction	9740/01	i) $S_1 = \frac{1}{2}, S_2 = \frac{2}{3}, S_3 = \frac{3}{4}, S_4 = \frac{4}{5}$ ii) $f(n) = \frac{1}{n+1}$	7
5	Curve Sketching	9740/01	i) $(0,0)$; $y=0$ ii) $(0,0.5)$; $y=2$ iii) $(0, \sqrt{2})$; $(0, -\sqrt{2})$; $y = \pm \frac{\sqrt{2}}{2}$	7
6	Functions	9740/01	i) $f^{-1}(x) = 1 - \sqrt{x-1}, 1 \leq x \leq 2$ ii) $gf(x) = \frac{x^2 - 2x + 4}{x^2 - 2x + 3}, 0 \leq x \leq 1$ iii) Range of $gf = \left[\frac{4}{3}, \frac{3}{2} \right]$	9
7	Differentiation	9740/01	i) y is a decreasing function. ii) $y = -x + 4a$ iii) Tangent does not meet the curve again	10
8	Curve Sketching & Inequalities	9740/01	a) Domain of $\cos^{-1} x: -1 \leq x \leq 1$ Range of $\cos^{-1} x: 0 \leq \cos^{-1} x \leq \pi$ Domain of $\tan^{-1} x: -\infty < x < \infty$ Range of $\tan^{-1} x: -\frac{\pi}{2} < \tan^{-1} x < \frac{\pi}{2}$ b) $x = \pm \frac{\sqrt{3}}{2}$ c) Reflection in the line $y = x$; $0 < x < \alpha$	11
9	Differential Equations	9740/01	ii) $y = \sqrt{2} \tan(\sqrt{2}x + \tan^{-1} \sqrt{2}) - x$ iii) $\frac{dx}{dz} = \frac{3}{2}$	9

10	Complex Numbers	9740/01	a) $z = 2 + 4i$; $w = -2 + 7i$ b) $z = \exp\{i(-\frac{\pi}{2})\}$, $\exp\{i(\frac{\pi}{6})\}$, $\exp\{i(\frac{5\pi}{6})\}$ $w = \exp\{i(\frac{\pi}{2})\}$, $\exp\{i(-\frac{\pi}{6})\}$, $\exp\{i(-\frac{5\pi}{6})\}$	9
11	Maclaurin's Series	9740/01	i) $\frac{dy}{dx} = -2(1 + y^2)$ ii) $1 - 4x + 8x^2 - \frac{64}{3}x^3$ iii) $a = 2$, $b = -\frac{1}{25}$, $c = \frac{1}{1250}$	10
12	Vectors	9740/01	i) $\frac{1}{3} \begin{pmatrix} -1 \\ 8 \\ 14 \end{pmatrix}$ ii) $r = \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1+a \\ 1-a \\ -2 \end{pmatrix}$ iii) $a = -3$; $b = 10$. iv) $a = 0.781$ or $a = 1.25$	14
Pure Maths				
1	Vectors	9740/02	$r = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$ $\overrightarrow{OC} = \begin{pmatrix} 1+3\lambda \\ 2-4\lambda \\ 1+\lambda \end{pmatrix}$ $AC : CB = 1 : 2$	7
2	Binomial Series	9740/02	$a=3$, $b=3$, $c=\frac{27}{16}$ $p=\frac{1}{16}$, $q=\frac{3}{2}$	11
3	Integration	9740/02	i) $\left(2, \frac{1}{2} \ln 2\right)$; $\left(5, \frac{1}{2} \ln 5\right)$ ii) $\frac{5}{2} \ln 5 - \ln 2 - \frac{3}{2}$;	10

			$\alpha = \frac{5}{2}, \quad \beta = -1, \quad \gamma = -\frac{3}{2}$	
			iii) 5.75 square units	
4	Complex Numbers	9740/02	i) $\frac{\sqrt{2}}{2}; \frac{\pi}{12}$ ii) $z = (2 + \sqrt{2}) + (1 + \sqrt{2})i$ or $z = (2 - \sqrt{2}) + (1 - \sqrt{2})i$	12
	Statistics			
5	P & C	9740/02	24; 12; 288	6
6	Binomial & Poisson Dist.	9740/02	i) $P(X = 18) = 0.0844$ ii) Variance = 882 iii) 0.955	5
7	Sampling Methods	9740/02	i) Stratified sampling is more representative of the entire population. Random sampling can't ensure that both genders are represented appropriately. ii) Biased or not random. The population of leaves beyond the student's reach has been excluded.	3
8	Binomial Dist.	9740/02	0.0412	3
9	C.L.T & Binomial	9740/02	i) 0.990 ii) 9.96×10^{-4} Assume $P(\text{success}) = 0.989868$ is a constant for each trial	6
10	Normal, Binomial & Poisson.	9740/02	i) (a) $P(X < 8) = 0.106$ (b) $P(X > 10.5) = 0.0304$ ii) Since 10.6% of the pickles are discarded because they are too short, it is impossible to reduce the total rejects to 5%. iii) $P(L > 8) = 0.00413$ iv) 0.290	10
11	Sampling, Testing	9740/02	i) To test $H_0: \mu = 3.6$ against $H_1: \mu < 3.6$ at 1 % level of significance ii) minimum $n = 780$ iii) $p\text{-value} = 0.0845 > 0.05$ Do not reject at 5% level of significance and conclude that there is insufficient evidence that the modification has decreased the mean diameter of the ball bearings at 5% level of significance. p value is the probability of	10

			wrongly rejecting the hypothesis that $\mu = 3.6$ i) $r = -0.952$ -For the depths sampled the moisture content decreases approximately linearly with an increase in the depth of the sample. -Since r is close to 1 , this implies that the regression line of m on x and x on m are almost identical or close to each other ii) 15.2 Reliable since the value of m is within the range of the given data. Not extrapolating. iii) $x = -0.351y + 20.7$	
12	Linear Correlation & Regression	9740/02	$\frac{37}{50}$ i) 0.172 ii) Least $n = 13$	9
13	Probability & GP	9740/02		9